

K for Kotlin

Kotlin is a relatively new programming language. Its development started in 2011 but it is gaining popularity. Kotlin is a statically typed programming language that runs over the Java virtual machine, and is widely used for Android app development. Kotlin is released under the Apache licence. Since Korn shell is Turing complete, it is worth an honourable mention.

L for Lisp

An electrical engineer might have chosen LabVIEW, a system-design platform and development environment, for the letter L. But I think Lisp, being the original functional programming language, should represent the letter L. Lisp is one of the oldest surviving high level programming languages. But do remember that it is not a single programming language; instead, there are a number of dialects like Clojure, Racket, Scheme, etc, for Lisp. Lua, a multi-paradigm programming language, is also worth mentioning here.

M for MATLAB

The letter M is for Maple, Magma, MATLAB, Maxima and Mathematica — the programming languages and mathematical software used to support serious mathematics rather than, say, finding the sum of two small integers and displaying the result on a Web page. Even though Maxima is open source software, I have chosen MATLAB, a proprietary commercial software. I believe the widespread use of MATLAB justifies its selection. M is also the letter that reminds us about machine languages and all the pioneers who wrote programs on punched cards.

N for NASM

N is another letter in this sequence whose removal will not affect the programming community much. Netwide Assembler (NASM), an

assembler and disassembler for the Intel x86 architecture, is my choice here. NASM is Turing complete; hence, it can be considered a programming language. It is released under the BSD licence.

O for Objective-C

Objective-C is an object-oriented programming language that has compilers implemented by both Clang and GCC. It is included as one of the Top Ten programming languages in the *Open Source For You* article mentioned earlier. Objective-C is supported by Apple for the development of macOS and iOS applications. Ocaml, a multi-paradigm programming language, and Octave, a programming language for numerical computations, are two runners-up for representing the letter O.

P for Python

P is another rich letter in our programming languages alphabet but yet again, the choice was simple — it is none other than Python. Python is an interpreted general-purpose programming language. In this era of artificial intelligence and machine learning, Python is the golden child in the family of programming languages. Applications in a wide variety of fields make Python absolutely important. But in this category, there are also other important programming languages worth mentioning. Pascal is an important procedural programming language developed in 1970. Perl is a general-purpose, interpreted, dynamic programming language often used for text processing. PHP is a general-purpose programming language most often used for Web development, while Prolog, a logic programming language associated with AI, is also worth mentioning.

Q for Q#

The choice for the letter Q is Q#, a programming language released by

Microsoft for developing quantum algorithms. Q# is the newest programming language included in our list, with the first release coming in December 2017. It is influenced by both C# and F#. Recently, we have heard that Google achieved quantum supremacy and hence will make quantum computers a reality in the near future, hopefully. So, it is possible that Q# or similar programming languages will play a pivotal role in the programming world in the not-so-distant future. The other two programming languages in this category worth mentioning are QtScript and QBasic.

R for R

Choosing the language for the alphabet 'R' was also relatively easy. R is yet another Top Ten ranked programming language that features in almost all rankings. R is a programming language and free software environment for statistical computing. Since Big Data and data mining are very popular nowadays, the popularity of R seems to be rising. Ruby, an interpreted general-purpose programming language, and Rust, a multi-paradigm system programming language, are also worth mentioning in this category.

S for Swift

S is another rich letter in the programming languages alphabet but here again the choice was simple; it is Swift. To my surprise I found out that there is also a parallel scripting language called Swift but the one I'm referring to is a general-purpose, multi-paradigm, programming language developed by Apple Inc. This, too, is ranked among the Top Ten in most rankings. Currently, Swift is released under Apache License 2.0 making it an open source programming language. There are a number of other programming languages in this category that are worth mentioning. Scala is a general-purpose functional